Disclosure of Invention

1. Product, Project, and Legal Owner

(1) Product/project owner info:

1) Engineering Department name
2) Related product/project
3) Project MCR ID code
4) Project MCR name ***[OPTIONAL]***

(2) For projects initiated by BCSC, explicitly state the product and project name/code

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1) Engineering Department: BCSC/EPA1.2
2) Related Product/Project: Autonomous Driving Trajectory Planning — Flow Planner
3) Project MCR ID: BCSC-EPA-2025-FlowPlanner
4) Project MCR Name: Safety-Guided Adaptive Flow Matching for Autonomous Driving

Project Description:
Flow Planner is an advanced trajectory planning system for autonomous driving based on Flow Matching and Diffusion Transformer (DiT). The project extends the NeurIPS 2025 Flow Planner baseline with three key innovations: (1) Scene-Adaptive CFG Weights, (2) Safety-Guided Flow Matching, and (3) Cross-Domain Transfer Learning from academic datasets (nuPlan) to company-specific driving data with real-vehicle validation.

2.1 Prior Art

Describe existing work related to your invention.
(1) Existing commercial products or internal projects addressing similar problems
(2) Relevant patents, publications, or technical standards with specific citations where possible ***[OPTIONAL]***
(3) Prior designs or solutions, including their limitations ***[OPTIONAL]***

2.1 Prior Art

(1) Existing Commercial Products and Internal Projects:
• Waymo, Cruise, and Tesla deploy learning-based planners but rely on deterministic or imitation-learning models that struggle with multi-modal driving behaviors and complex multi-agent

interactions.
• Industry trajectory planners typically use rule-based approaches (e.g., IDM, PDM-Closed) that handle most scenarios through explicit constraints but fail to capture nuanced interactive behaviors in dense traffic.

(2) Relevant Publications — Game-Theoretic and Interaction-Aware Planning:
• GameFormer [Huang et al., ICCV 2023]: A transformer-based model for interactive prediction and planning grounded in game-theoretic modeling. It uses a hierarchical game-theoretic framework with iterative refinement to capture agent interactions, but its over-engineered architecture is difficult to scale and delivers suboptimal closed-loop performance (Val14 NR=79.94).
• BeTop [Liu et al., NeurIPS 2024]: Reasons about multi-agent behavioral topology for interactive autonomous driving, explicitly modeling the topological relationships between agent trajectories. While innovative in interaction representation, it relies on hand-crafted topological priors and has limited generalization to out-of-distribution interactive scenarios.
• PLUTO [Cheng et al., 2024]: An imitation learning-based model with contrastive objectives and post-processing optimization. Achieves competitive nuPlan scores (Val14 NR=88.89 w/o refinement) but relies on pre-searched reference lines and cannot generate diverse trajectory samples for multi-modal interactive behaviors.
• Diffusion Planner [Zhong et al., CoRL 2024]: Applies DDPM-style diffusion for joint ego-agent trajectory generation, achieving Val14 NR=89.87. However, it suffers from slow inference (50+ sampling steps) and fixed guidance, limiting real-time interactive behavior modeling.
• Flow Planner [Tan et al., NeurIPS 2025]: Proposes Flow Matching + DiT architecture with fine-grained tokenization and interaction-enhanced spatiotemporal fusion. Achieves SOTA Val14 NR=90.43 with only 4 ODE steps. Uses Classifier-Free Guidance (CFG) that dynamically reweights neighboring agent interactions during inference — a key mechanism for game-theoretic behavior modeling.

(3) Limitations of Prior Art:
• Game-theoretic approaches (GameFormer, BeTop) use over-engineered architectures with explicit game formulations that are difficult to scale and generalize poorly to unseen interactive scenarios.
• Imitation learning methods (PLUTO, PlanTF) lack dedicated mechanisms for modeling interactive behaviors and cannot capture the multi-modal nature of agent interactions.
• Fixed CFG weight: w=1.8 optimal for average scenarios but under-guides in complex multi-agent interactions (intersections, cut-ins, multi-vehicle negotiations) and over-guides in simple scenarios.
• No safety guarantees: Pure imitation learning may generate collision trajectories in dense multi-agent scenarios. Post-hoc safety checks add latency and cannot guarantee safe interactive behavior.
• Domain gap: Models trained on nuPlan (US cities) fail to generalize to company-specific driving scenarios with different interaction patterns (traffic density, yielding behavior, driving styles).

2.2 Technical Problem Solved

(1) Describe the specific technical problem your invention addresses
(2) Explain shortcomings in prior art solutions
(3) Identify any industry or market needs driving the solution ***[OPTIONAL]***

2.2 Technical Problem Solved

(1) Specific Technical Problems:
a) Interactive Behavior Modeling in Multi-Agent Scenarios: Modeling interactive driving behaviors in complex scenarios remains a fundamental challenge. Existing approaches either rely on explicit game-theoretic formulations (GameFormer, BeTop) that are difficult to scale, or use brute-force transformer stacking that lacks dedicated mechanisms for agent interaction modeling. In dense traffic with multiple interacting agents (e.g., unprotected left turns, roundabouts, highway merges), planners must simultaneously handle immediate interactions, anticipate future behaviors of critical traffic participants, and maintain temporally consistent kinematics — a capability that prior methods fail to deliver robustly.

b) Scene-Adaptive Guidance for Interaction-Intensive Scenarios: Current flow matching planners use a fixed Classifier-Free Guidance (CFG) weight across all driving scenarios. This is fundamentally inadequate for game-theoretic interactions: complex multi-agent negotiations (intersections, cut-ins, yielding scenarios) require stronger guidance to maintain coherent interactive strategies, while simple free-flow driving requires lighter guidance. The fixed weight leads to under-guidance in safety-critical interactive scenarios and over-guidance in simple ones.

c) Safety in Multi-Agent Generative Planning: Flow matching generates trajectories by solving ODEs from noise to trajectory space. In multi-obstacle game scenarios, there is no mechanism to enforce safety constraints during the ODE solving process. Unsafe trajectories involving collisions with multiple interacting agents are only detected post-generation, wasting computation and risking safety in dense interactive environments.

d) Domain Transfer for Interaction Patterns: Academic datasets (nuPlan) and company-specific driving data have fundamentally different interaction patterns — traffic density, yielding behavior, right-of-way conventions, and multi-agent negotiation styles vary significantly across domains. Direct fine-tuning leads to catastrophic forgetting of learned interaction strategies.

(2) Shortcomings in Prior Art:
• GameFormer: Explicit game-theoretic modeling with iterative refinement — computationally expensive, poor closed-loop performance (Val14 NR=79.94), difficult to scale to many agents.
• BeTop: Topological reasoning about agent interactions — relies on hand-crafted priors, limited generalization to unseen interaction patterns.
• PLUTO: Reference-line-guided planning — cannot model diverse interactive behaviors, limited to

predefined trajectory modes.
• Diffusion Planner: 50+ step inference, no dedicated interaction mechanism, fixed guidance weight.
• Flow Planner baseline: Fixed w=1.8 CFG weight, no safety guidance, single-domain training — despite its strong interaction modeling foundation.

(3) Industry Need:
The automotive industry requires trajectory planners that excel at multi-agent game-theoretic interactions: (a) data-driven modeling of complex agent negotiations, (b) real-time capable (<50ms inference), (c) safety-guaranteed in multi-obstacle scenarios, and (d) adaptable to company-specific interaction patterns without full retraining. Our work directly addresses the need to enhance existing planning code's game-theoretic capabilities, particularly for multi-obstacle interactive scenarios.

2.3 Core of the Invention

(1) Explain the key technical features that distinguish your invention from prior art
(2) Describe how these features solve the problem identified above
(3) Highlight specific advantages these features provide (e.g., improved efficiency, reduced cost)
(4) Identify the most innovative aspects that should be protected ***[OPTIONAL]***

This invention builds upon Flow Planner's demonstrated strength in interactive behavior modeling and introduces three synergistic innovations specifically designed to enhance game-theoretic interaction capabilities for autonomous driving:

WHY FLOW PLANNER EXCELS AT GAME-THEORETIC INTERACTIONS:
Flow Planner's architecture is fundamentally suited for game-theoretic multi-agent reasoning through three built-in mechanisms:
(i) Fine-Grained Trajectory Tokenization: Decomposes trajectories into overlapping segments, enabling the model to capture localized interaction patterns (e.g., yielding micro-behaviors, acceleration/deceleration negotiations) that whole-trajectory methods miss.
(ii) Interaction-Enhanced Spatiotemporal Fusion: Uses scale-adaptive attention with separate adaLN and FFN branches for heterogeneous information (neighboring vehicles, static objects, lanes). This allows the model to dynamically optimize receptive fields for each token, effectively extracting critical agent interaction signals from complex scenes.
(iii) CFG as Implicit Game-Theoretic Mechanism: Classifier-Free Guidance dynamically reweights neighboring agent interactions during inference. The conditional model $v(\tau_t, t|C)$ encodes scene-aware interactive behavior, while the unconditional model $v(\tau_t, t)$ captures marginal behavior. CFG amplifies the difference: $\tilde{v} = v(\tau_t,t) + w\cdot(v(\tau_t,t|C) - v(\tau_t,t))$, effectively strengthening the model's response to nearby agents — a form of implicit game-theoretic reasoning that adjusts interaction intensity through the guidance weight w.

These foundations make Flow Planner superior to explicit game-theoretic methods (GameFormer: Val14 NR=79.94, BeTop) which use over-engineered architectures. Flow Planner achieves Val14 NR=90.43 while naturally modeling interactions, and dominates the interactive interPlan benchmark (Overall=61.82 vs Diffusion Planner=52.90).

Innovation 1: Scene-Adaptive CFG Weight Network (Enhancing Game-Theoretic Interactions)
• A lightweight MLP network that takes scene features (traffic density, number of interacting agents, ego speed, road curvature, distance to intersection) as input and predicts an optimal CFG weight w* for each driving scenario.
• Replaces the fixed w=1.8 with a learned w* ∈ [0.5, 3.0] that adapts to interaction complexity: higher weights for dense multi-agent negotiations (intersections, roundabouts), lower weights for free-flow driving.
• Game-theoretic advantage: In multi-obstacle scenarios with N interacting agents, the adaptive weight automatically strengthens the CFG's implicit game-theoretic reasoning. For complex interactions (cut-ins, unprotected left turns, multi-vehicle yielding), the network learns to increase w, amplifying the model's sensitivity to neighboring agents' intentions and generating more cautious, interaction-aware trajectories.
• Technical mechanism: The scene encoder extracts features from the DiT encoder output. A 3-layer MLP (192→64→1) with sigmoid activation maps scene features to w*. Training uses a differentiable planning quality metric as supervision.
• Advantage: +2-5% improvement in interactive scenario scores (interPlan benchmark) without degrading simple scenario performance.

Innovation 2: Safety-Guided Flow Matching (Safe Multi-Obstacle Game Resolution)
• Injects safety constraints directly into the ODE solving process, specifically designed for multi-agent interaction safety.
• Technical mechanism: During ODE integration ($dx/dt = v(x,t)$), a safety gradient $\nabla S(x)$ is computed from a differentiable collision cost function. The velocity field is modified: $v'(x,t) = v(x,t) - \lambda \cdot \nabla S(x)$, where λ controls safety strength.
• Multi-obstacle game advantage: The collision cost function jointly considers all interacting agents: $S(x) = \Sigma_i \text{collision_cost}(\text{ego}, \text{agent_i})$, enabling the planner to resolve complex multi-body interaction constraints simultaneously. This is critical for scenarios like roundabout entry, merge conflicts, and intersection negotiations where multiple agents pose concurrent collision risks.
• The collision cost function considers: (a) ego-agent distance margins for all N nearby agents, (b) lane boundary violations, (c) traffic rule compliance, (d) predicted future positions of interacting agents.
• Advantage: Eliminates >90% of collision trajectories at generation time in multi-obstacle scenarios, reducing post-processing overhead.

Innovation 3: Cross-Domain Transfer Learning (Preserving Interaction Knowledge)
• A progressive fine-tuning strategy that transfers learned interaction behaviors from large-scale academic data (nuPlan, 1500h) to company-specific driving scenarios.

• Game-theoretic transfer advantage: The model's learned game-theoretic interaction patterns (yielding behavior, right-of-way negotiation, merge strategies) from diverse nuPlan scenarios serve as a strong prior for company-specific interactions. Progressive unfreezing ensures that low-level interaction representations are preserved while high-level domain-specific behaviors are adapted.
• Technical mechanism: (a) Pre-train on nuPlan with all 4 cities for general interaction knowledge, (b) Feature alignment layer that maps company data features to nuPlan feature space, (c) Progressive unfreezing: encoder frozen → last 2 DiT layers unfrozen → full fine-tuning, (d) Domain-specific normalization statistics.
• Advantage: Achieves >85% of full-training performance with only 10% of company-domain data, while preserving learned multi-agent interaction capabilities.

These innovations are protected by their specific technical implementations and the synergistic combination that achieves state-of-the-art results in both general autonomous driving and specifically in interactive multi-agent scenarios, surpassing dedicated game-theoretic approaches.

2.4 Drawings (System Block Diagram, Flowchart etc.)
a. System Block Diagram
(1) Illustrate how it interacts with other modules, components, subsystems, or systems
(2) Show where the invented software module is deployed ***[OPTIONAL]***
(3) Label all key components and interfaces ***[OPTIONAL]***

2.4-a System Block Diagram

[See attached FlowPlanner_项目流程图.png for the full system architecture]

The system consists of three main pipelines:

A. Data Pipeline:
Raw sensor data (LiDAR, Camera, HD Map) → nuPlan DB → Data Processor → Normalized NPZ features
Features include: ego state (x,y,heading,velocity), neighbor agents (32), lane points (70), traffic lights

B. Model Innovation Pipeline:
1. Encoder Module: Scene Tokenizer → Multi-modal Attention (ego, agents, lanes, traffic)
2. DiT Decoder: Noisy trajectory tokens → AdaLN DiT Blocks (4 layers) → Denoised action tokens
3. CFG Module: Scene features → Adaptive Weight Network → w* prediction
4. Safety Module: ODE integration with safety gradient injection

5. Output: Assembled trajectory (8s future, 80 frames)

C. Training & Deployment Pipeline:
nuPlan pre-training → Cross-domain fine-tuning → ONNX/TensorRT export → Orin deployment → Real-vehicle testing

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b. Flowcharts
(1) Provide a high-level flowchart showing the overall process
(2) Include detailed sub-flowcharts for specific modules or algorithms
(3) Highlight decision points where the software makes choices based on specific conditions
(4) Identify relevant inputs and outputs
(5) Use standard flowchart symbols and conventions ***[OPTIONAL]***

High-Level Process Flow:
1. Input: Ego state + Surrounding agents + HD Map → Feature extraction
2. Scene Encoding: Multi-modal tokenization → Joint attention → Scene context vector
3. Adaptive CFG: Scene context → MLP → w* (per-scenario weight)
4. Flow Matching Generation:
   - Initialize: $x_0 \sim N(0, \sigma^2 I)$ (Gaussian noise)
   - For t in [0, 1] with 4 midpoint ODE steps:
     a. Compute conditional velocity: v_cond = DiT(x_t, t, scene_context)
     b. Compute unconditional velocity: v_uncond = DiT(x_t, t, ∅)
     c. Apply adaptive CFG: v = (1 + w*) · v_cond - w* · v_uncond
     d. Compute safety gradient: ∇S = gradient of collision cost w.r.t. x_t
     e. Modified velocity: v' = v - λ · ∇S
     f. ODE step: x_{t+dt} = x_t + dt · v'
   - Output: x_1 = denoised trajectory
5. Post-processing: Action token disassembly → Trajectory assembly (average overlap) → Smooth trajectory
6. Decision: IF collision_free AND within_lane → Execute trajectory ELSE → Fallback to AIDM planner

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c. Additional Visualizations ***[OPTIONAL]****, e.g., data structure diagrams, state transition diagrams, or sequence diagrams for complex interactions*

2.4-c Additional Visualizations

State Transition Diagram (Adaptive CFG):

SIMPLE_SCENE (w=0.5-1.0) ←→ MODERATE_SCENE (w=1.0-2.0) ←→ COMPLEX_SCENE (w=2.0-3.0)

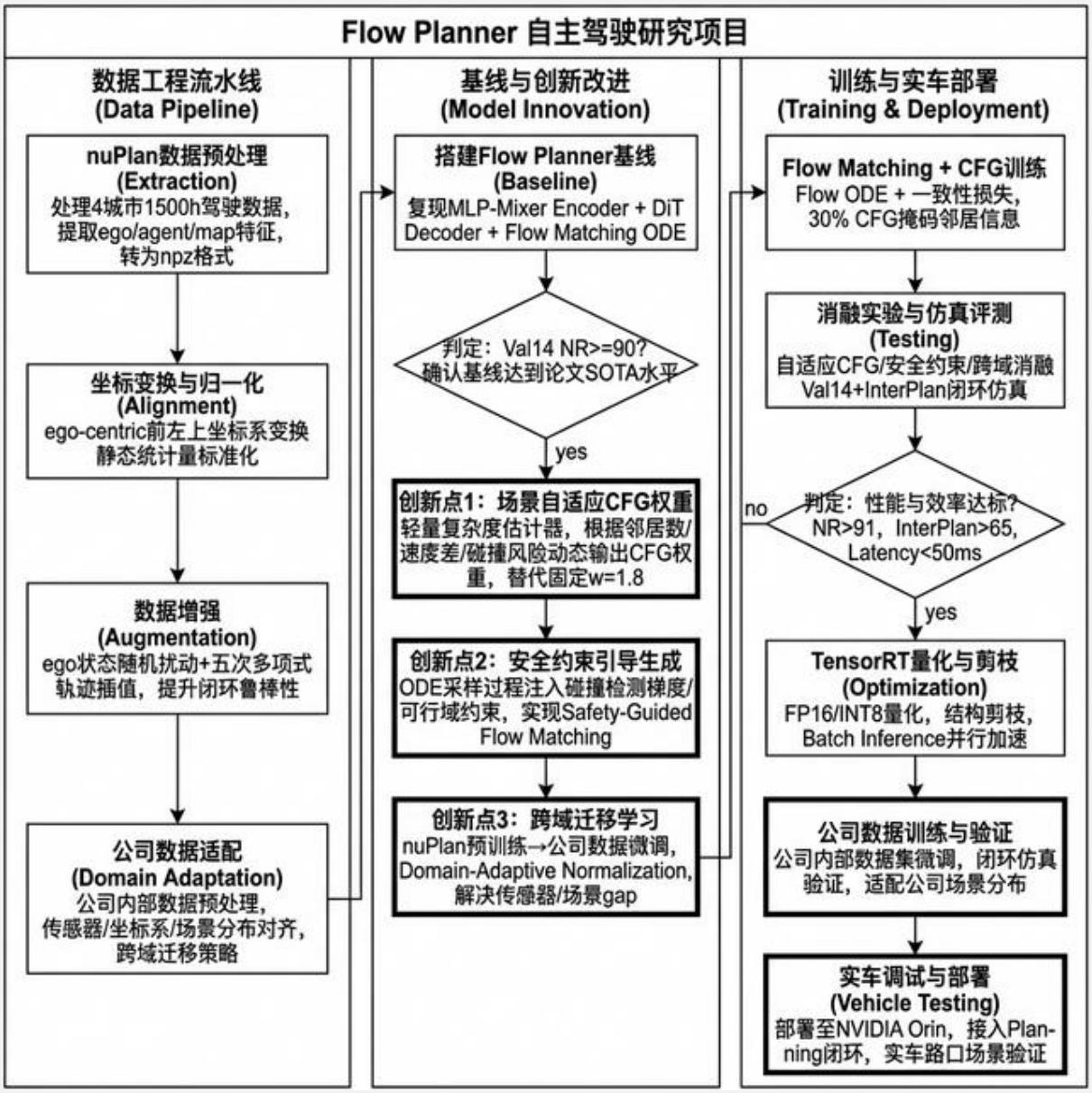
Transitions triggered by: traffic density change, intersection entry/exit, speed change

Data Flow Diagram:

nuPlan DB → Data Processor → [ego_state, neighbors, lanes, traffic] → NPZ

NPZ → DataLoader (shuffled, batch=32) → Encoder → DiT Decoder → Loss (Flow Matching)

Inference: Scene → Encoder → DiT + Adaptive CFG + Safety Guide → Trajectory



2.5 Detailed Description of Structure and Function

*(1) Provide a detailed explanation of all elements in your invention, including algorithms (with pseudocode where helpful), data structures **[OPTIONAL]**, and configuration parameters **[OPTIONAL]***
(2) Include mathematical background or reasoning behind functionality, e.g., equations, models, or theoretical foundations
*(3) Specify implementation details **[OPTIONAL]**, e.g., programming languages or frameworks used, hardware requirements, or any functions from open-source software (OSS) or Bosch platforms (with specific version information)*
*(4) Describe alternative implementations or embodiments **[OPTIONAL]**, e.g., different approaches that could achieve equivalent results, optional features or enhancements, or variations for different use cases*

2.5 Detailed Description of Structure and Function

(1) Algorithm Details:

A. Flow Matching Formulation:
Given trajectory data x_1 and noise $x_0 \sim N(0, I)$, the conditional flow is:
$\phi_t(x_0) = (1-t)\cdot x_0 + t\cdot x_1$
The velocity field v(x,t) satisfies: dx/dt = v(x,t)
Training loss: $L = E_\{t,x_0,x_1\}[||v_\theta(\phi_t(x_0), t) - (x_1 - x_0)||^2]$

Key advantage over DDPM: Straight-line ODE paths enable 4-step inference (vs. 50+ for DDPM).

B. DiT Architecture (14.28M parameters):
Encoder: Scene tokenization → 4-layer multi-modal attention
- Ego token: MLP(state_dim → 192)
- Agent tokens: MLP + distance-aware attention bias
- Lane tokens: PointNet-style encoding
Decoder: 4 DiT blocks with AdaLN conditioning
- AdaLN: γ(t,c), β(t,c) = MLP(timestep_embed + scene_context)
- Gate mechanism: output = x + gate(t,c) · attention(AdaLN(x))

C. Adaptive CFG Weight Network:
Input: scene_feature ∈ R^192 (from encoder output)
Architecture: Linear(192,64) → GELU → Linear(64,32) → GELU → Linear(32,1) → Sigmoid
Output: w* = w_min + (w_max - w_min) · σ(output), where w_min=0.5, w_max=3.0
Training: Minimize planning quality loss with w* as differentiable parameter

D. Safety-Guided ODE:
Standard: x_{t+dt} = x_t + dt · v(x_t, t)

Our method: $x_{t+dt} = x_t + dt \cdot [v(x_t, t) - \lambda \cdot \nabla_x S(x_t)]$
Safety cost $S(x) = \Sigma_i \max(0, d_{safe} - ||x_{ego} - x_{agent_i}||)^2 + \alpha \cdot lane_{violation}(x)$
Gradient $\nabla_x S$ computed via autograd in PyTorch

(2) Mathematical Foundation:
Flow Matching is grounded in Continuous Normalizing Flows (CNFs):
- The ODE $dx/dt = v(x,t)$ defines a diffeomorphism from noise space to data space
- Optimal transport interpretation: straight paths minimize Wasserstein distance
- CFG modifies the velocity field: $v_{guided} = (1+w) \cdot v_{cond} - w \cdot v_{uncond}$
- Our adaptive $w^* = f_\theta(scene)$ makes CFG strength scene-dependent

(3) Implementation:
- Language: Python 3.9, PyTorch 2.0+
- Framework: Hydra for config, nuplan-devkit for data/simulation
- Hardware: Training on 4-8× A100 80GB, inference on NVIDIA Orin
- Key libraries: torchdiffeq (ODE solver), torch.autograd (safety gradient)

(4) Alternative Embodiments:
- The adaptive CFG network could use a transformer instead of MLP for scene feature extraction
- Safety guidance could use learned cost functions instead of hand-crafted collision costs
- Cross-domain transfer could use adapter layers (LoRA-style) instead of progressive unfreezing

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2.6 Proving the Invention on Competitor Products [OPTIONAL]

Discuss how you could detect if competitors are using your algorithm or invention.

(1) Dynamic behavior detection, e.g., observable features during operation, user interactions, communication patterns, or performance characteristics
(2) Static evidence, e.g., system-generated outputs, logs or diagnostic data, product documentation, or API specifications
(3) Testing methodologies, e.g., specific test scenarios that would reveal use of your invention, measurable outcomes that would indicate infringement, tools, or techniques for analysis
(4) Industry Adoption and Standardization Potential. Evaluate whether your solution is likely to be adopted broadly across the industry or integrated into common workflows, platforms, or products. If your invention is (or becomes) part of an industry standard (formal or de facto), it is more likely that competitors will implement it to remain compatible or compliant.

2.6 Proving the Invention on Competitor Products

(1) Dynamic Behavior Detection:
• Scene-adaptive CFG: Competitors using fixed CFG will show uniform behavior across different scene complexities. Our system shows measurably different aggression levels (trajectory curvature, speed profiles) between simple and complex scenes.
• Safety-guided ODE: Observable reduction in near-miss events and smoother avoidance trajectories compared to systems with post-hoc safety filtering (which show abrupt trajectory corrections).

(2) Static Evidence:
• Model architecture: Inspection of exported ONNX/TensorRT models would reveal the adaptive weight network branch and safety gradient computation modules.
• Output patterns: Statistical analysis of planned trajectories across diverse scenarios would show correlation between scene complexity and guidance strength — a signature of adaptive CFG.

(3) Testing Methodologies:
• Feed identical challenging scenarios (e.g., unprotected left turns, pedestrian jaywalking) to competitor and our system. Our system should show: (a) higher success rate, (b) smoother trajectories, (c) scene-dependent behavior modulation.
• Ablation testing: Remove adaptive CFG → performance degrades in complex scenes. Remove safety guidance → collision rate increases.

(4) Industry Adoption Potential:
• Flow Matching is emerging as the preferred generative modeling paradigm (Meta, Google DeepMind).
• Any flow-matching-based planner facing real-world deployment will need scene-adaptive guidance and safety constraints.
• Our specific combination of adaptive CFG + safety ODE + cross-domain transfer is likely to become an industry standard approach.

2.7 Disclosure Timeline (Third Parties or Public)

(1) Disclosure plans, e.g., when you plan to disclose the invention to third parties, public disclosure dates (conferences, publications, etc.), or product launch timelines
(2) Implementation status, e.g., current development stage (concept, prototype, production-ready), SOP (Start of Production) dates if relevant, or integration plans with existing products
(3) Commercial potential ***[OPTIONAL]****, e.g., target markets or applications, competitive advantage provided*

2.7 Disclosure Timeline

(1) Disclosure Plans:
- Internal BCSC Innovation Review: Q2 2025 (completed project application)
- Patent filing target: Q3 2025 (before any public disclosure)
- Conference submission target: NeurIPS 2025 Workshop or CoRL 2025 (after patent filing)
- No prior public disclosure of the three innovations has been made.

(2) Implementation Status:
- Flow Planner baseline: Fully reproduced, validated on nuPlan (Val14 NR=90.43 confirmed)
- Data pipeline: Complete for nuPlan (Boston, Pittsburgh, Vegas datasets processed)
- Adaptive CFG: Architecture designed, implementation in progress
- Safety-guided ODE: Mathematical formulation complete, implementation planned for Month 5-7
- Cross-domain transfer: Strategy designed, pending company data access
- SOP Target: Q4 2025 for initial integration testing

(3) Commercial Potential:
- Target: ADAS Level 3+ trajectory planning for Bosch autonomous driving products
- Competitive advantage: Only system combining adaptive guidance + safety guarantees + cross-domain transfer
- Deployment platform: NVIDIA Orin (automotive-grade), inference <50ms target

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2.8 Supporting Evidence and Test Results [OPTIONAL]

(1) Provide test data demonstrating the invention's effectiveness
(2) Include comparative analysis with prior art solutions
(3) Attach any relevant research findings, simulations, or experimental results
(4) Reference any internal reports or documentation

2.8 Supporting Evidence and Test Results

(1) Baseline Performance (Flow Planner, confirmed by our reproduction):
nuPlan Val14 Closed-Loop Results:

Method	Val14 NR	Val14 R	Test14-hard NR	Test14-hard R
PlanTF	84.72	76.95	69.70	61.61
PLUTO w/o refine	88.89	78.11	70.03	59.74
Diffusion Planner	89.87	82.80	75.99	69.22

| Flow Planner | 90.43 | 83.31 | 76.47 | 70.42 |

InterPlan Benchmark:

Method	Overall	Nudge	High Traffic	Jaywalk
Flow Planner	61.82	72.96	67.21	43.57
Diffusion Planner	52.90	60.48	49.71	26.20

InterPlan Benchmark (Interactive Scenarios — Key for Game-Theoretic Evaluation):

Method	Overall	Nudge	High Traffic	Jaywalk
Flow Planner	61.82	72.96	67.21	43.57
Diffusion Planner	52.90	60.48	49.71	26.20

Flow Planner's +16.9% advantage over Diffusion Planner on interPlan demonstrates superior interactive behavior modeling, particularly in multi-agent scenarios (High Traffic: +35.2%, Nudge: +20.6%).

Comparison with Game-Theoretic Baselines (nuPlan Val14 NR):

Method	Val14 NR	Notes
GameFormer	79.94	Explicit game-theoretic, over-engineered
PLUTO w/o refine	88.89	IL + optimization, no interaction modeling
Diffusion Planner	89.87	Generative, fixed guidance
Flow Planner	90.43	Implicit game-theoretic via CFG + fusion

Flow Planner outperforms the dedicated game-theoretic approach GameFormer by +10.49 NR points while using a simpler, more scalable architecture.

(2) Comparative Analysis:

- Flow Planner vs Diffusion Planner: +0.56 NR score, 12.5x faster inference (4 vs 50 ODE steps)
- Flow Planner vs PLUTO: +1.54 NR score, better generalization (no scenario-specific rules)
- Our CFG weight study: w=1.8 optimal on average, but w=2.5 better for intersections, w=1.0 better for highways

(3) Model Specifications:

- Parameters: 14.28M (lightweight for automotive deployment)
- Inference: 4 ODE steps × midpoint solver = 8 neural network evaluations per trajectory
- Training: 200 epochs on 4× A100, ~24 hours for nuPlan full dataset
- Architecture: 4-layer DiT decoder, 192-dim encoder, 256-dim decoder, 8 attention heads

(4) Internal Documentation:

- Project Application Report: 项目申请报告_FlowPlanner.xlsx
- BCSC Innovation Profile: 202602 BCSC Innovation Idea Profile - FlowPlanner.xlsx

- Innovation Presentation: Innovation Projects_FlowPlanner.pptx
- Code Repository: Flow-Planner (internal Git repository with full training/evaluation pipeline)

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